$$c^- = \begin{bmatrix} c_1^- \\ \vdots \\ c_{N-1}^- \end{bmatrix}, \quad c^+ = \begin{bmatrix} c_1^+ \\ \vdots \\ c_{N-1}^+ \end{bmatrix}$$

$$\dot{c}^- = A^- c^- + B^- I$$

$$\downarrow$$

$$c_{k+1}^- = c_k^- + \Delta t \left(A^- c_k^- + B^- I_k \right) + \omega_k$$

$$= \underbrace{\left(I + \Delta t \cdot A^- \right)}_{A^{-\prime}} c_k^- + \underbrace{\Delta t \cdot B^-}_{B^{-\prime}} I_k + \omega_k$$

RK4

$$c_{k+1}^- = c_k^- + \Delta t \cdot \frac{k_1 + 2k_2 + 2k_3 + k_4}{6} + \omega_k$$

where

$$k_1 = A^- c_k^- + B^- I_k$$

$$k_2 = A^- \left(c_k^- + k_1 \frac{\Delta t}{2} \right) + B^- I_k$$

$$k_3 = A^- \left(c_k^- + k_2 \frac{\Delta t}{2} \right) + B^- I_k$$

$$k_4 = A^- \left(c_k^- + k_3 \Delta t \right) + B^- I_k$$

$$\ldots = A^{-\prime} c_k^- + B^{-\prime} I + \omega_k$$

$$V_k = \underbrace{\frac{RT}{\alpha F} \cdot \sinh^{-1}\left(\frac{I_k}{2 a^+ L^+ i_{0,k}^+} \right)}_{\bar{\eta}^+} - \underbrace{\frac{RT}{\alpha F} \sinh^{-1}\left(\frac{I_k}{2 a^- L^- i_{0,k}^-} \right)}_{\bar{\eta}^-} + U^+(c_{N,k}^+) - U^-(c_{1,k}^-)$$

$$- \left(\frac{R_S^+}{a^+ L^+} + \frac{R_S^-}{a^- L^-} \right) I_k + \frac{L^+ + 2L^{sep} + L^-}{2\kappa} \cdot I_k + k_{conc,k} \left[\ln c_{e,0^+,k} - \ln c_{e,0^-,k} \right]$$

$$= g\left(c_k^-, c_k^+, I_k \right) = g\left(c_k, I_k \right)$$

where $\bar{i}_0^{\pm} = k^{\pm}\left[c_{N,k}^{\pm}\right]^{\alpha_c}\left[c_{e,k}\cdot\left(c_{max}^{\pm} - c_{N,k}^{\pm}\right)\right]^{\alpha_a}$

discrete

$$H_k = \left.\frac{\partial g(c_k, I_k)}{\partial c_k}\right|_{c_k=\hat{c}_k}$$

$$= \left[\frac{\partial g}{\partial c_1^-}, \frac{\partial g}{\partial c_2^-}, \cdots, \frac{\partial g}{\partial c_{N-2}^-}, \frac{\partial g}{\partial c_{N-1}^-}, \frac{\partial g}{\partial c_1^+}, \frac{\partial g}{\partial c_2^+}, \cdots, \frac{\partial g}{\partial c_{N-2}^+}, \frac{\partial g}{\partial c_{N-1}^+}\right]$$

$$= \left[0, 0, \cdots, \frac{\partial g}{\partial c_N^-}\cdot\frac{\partial c_N^-}{\partial c_{N-2}^-}, \frac{\partial g}{\partial c_N^-}\cdot\frac{\partial c_N^-}{\partial c_{N-1}^-},\right.$$

$$\left.0, 0, \cdots, \frac{\partial g}{\partial c_N^+}\cdot\frac{\partial c_N^+}{\partial c_{N-2}^+}, \frac{\partial g}{\partial c_N^+}\cdot\frac{\partial c_N^+}{\partial c_{N-1}^+}\right]$$

continuous

$$H(t) = \frac{\partial g(c(t), I(t))}{\partial c}$$

$$3c_N^{\pm} - 4c_{N-1}^{\pm} + c_{N-2}^{\pm} = \pm\frac{2\Delta\bar{r}R_s^{\pm}}{D_s^{\pm}Fa^{\pm}A^{\pm}L^{\pm}}I$$

$$\Rightarrow \frac{\partial c_N}{\partial c_{N-1}} = \frac{4}{3} \quad \frac{\partial c_N}{\partial c_{N-2}} = -\frac{1}{3}$$

(when $g = V = g(c^+, c^-, u)$)

$$\frac{\partial g}{\partial c_N^{\pm}} = \pm\frac{\partial \eta^{\pm}}{\partial c_N^{\pm}} \pm \frac{\partial U^{\pm}}{\partial c_N^{\pm}} = \pm\frac{\partial \eta^-}{\partial \bar{i}_0}\cdot\frac{\partial \bar{i}_0^-}{\partial c_N^-} \pm \frac{\partial U^-}{\partial c_N^-}$$

$$\eta^{\pm} = \frac{RT}{\alpha F}\sinh^{-1}\left(X_k\right)$$

$$\frac{\partial \eta^{\pm}}{\partial i_0} = \frac{RT}{\alpha F} \frac{\partial \left(\sinh^{-1}\left(X_k^{\pm}\right)\right)}{\partial X_k^{\pm}} \cdot \frac{\partial X_k^{\pm}}{\partial i_{0,k}}$$

where $X_k^{\pm} = \frac{\mp I_k}{2 a^- L^- i_{0,k}^-}$

$$= \pm \frac{RT}{\alpha F} \cdot \frac{1}{\sqrt{X_k^2 + 1}} \cdot \frac{I_k}{2 a^- L^-} \left(\frac{1}{i_{0,k}^2} \right)$$

$$= \pm \frac{RT \cdot I_k}{2 \alpha F a^- L^- \cdot i_{0,k}^2 \sqrt{X^2 + 1}}$$

$$\frac{\partial X_k^{\pm}}{\partial i_{0,k}^-} = \frac{\partial \left(\mp \frac{I_k}{2 a^- L^-} \cdot \left(i_0^k\right)^{-1} \right)}{\partial i_{0,k}^-}$$

$$= \mp \frac{I_k}{2 a^- L^-} \cdot (-1) \left(i_0^k\right)^{-2}$$

$$= \pm \frac{I_k}{2 a^- L^- \left(i_0^k\right)^2}$$

$$\frac{\partial i_0^{\pm}}{\partial C_N} = \frac{\partial \left(k \left(C_{N,k} \cdot C_{e,k} \left(C_{max} - C_{N,k} \right) \right)^{\frac{1}{2}} \right)}{\partial C_N}$$

when $\alpha_c = \alpha_a = \frac{1}{2}$

$$= \frac{k \left(C_{e,k}\right)^{\frac{1}{2}} \partial \left(\left[C_{N,k} \left(C_{max} - C_{N,k} \right) \right]^{\frac{1}{2}} \right)}{\partial C_{N,k}}$$

$$= k \left(C_e\right)^{\frac{1}{2}} \cdot \frac{1}{2} \cdot X^{-\frac{1}{2}} \cdot \frac{\partial X}{\partial C_N}$$

where $X = C_N \left(C_{max} - C_N \right)$

$$\frac{\partial X}{\partial C_N} = \left(C_{max} - C_N \right) - \left(C_N \right) = C_{max} - 2 C_N$$

Sign check

	I	i_0^+	i_0^-	X^+	X^-	η^+	η^-
discharge	+	+	+	−	+	−	+
charge	−	+	+	+	−	+	−

	$\frac{\partial \eta^+}{\partial i_0^+}$	$\frac{\partial \eta^-}{\partial i_0^-}$	$\frac{\partial i_0^+}{\partial c_N^+}$		$\frac{\partial i_0^-}{\partial c_N^-}$	
discharge	+	−	− when $c_N^+ > \frac{1}{2} c_{max}^+$	+ when $c_N^+ < \frac{1}{2} c_{max}$	− when $c_N^- > \frac{1}{2} c_{max}^-$	+ when $c_N^+ < \frac{1}{2} c_{max}$
charge	−	+	− 〃	+ 〃	− 〃	+ 〃

$$\dot{c}^+ = A^+ c^+ + B^+ u + \underset{[mol/m^3]}{w}$$

$$\dot{c}^- = A^- c^- + B^- u - w$$

when $y = \begin{bmatrix} V \\ n_{Li} \end{bmatrix} = g(c^+, c^-, u) + v$

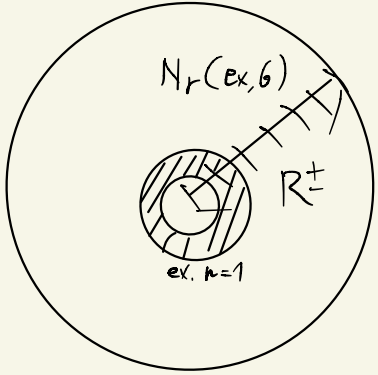
$$\Sigma_w = \underset{[(mol/m^3)^2]}{} \begin{bmatrix} \overbrace{\begin{matrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \left(\frac{0.04}{3600} c^+_{max}\right)^2 \end{matrix}}^{14} & 0 \\ 0 & \overbrace{\begin{matrix} 0 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \left(\frac{-0.04}{3600} c^-_{max}\right)^2 \end{matrix}}^{14} \end{bmatrix} \begin{matrix} \} 14 \\ \\ \} 14 \end{matrix}$$

$$\Sigma_v = \begin{bmatrix} \underset{[(V)^2]}{0.005} & 0 \\ 0 & 0 \end{bmatrix}$$

$\left(\Sigma_v = [0.005] \text{ when } y = V \right)$

where $g = \begin{bmatrix} V \\ n_{Li} \end{bmatrix} = \begin{bmatrix} g_1(c^+, c^-, u) \\ g_2(c^+, c^-, u) \end{bmatrix}$

$$H = \begin{bmatrix} 0, \cdots, \frac{\partial g_1}{\partial c^+_{N-2}}, \frac{\partial g_1}{\partial c^+_{N-1}}, 0, \cdots, \frac{\partial g_1}{\partial c^-_{N-2}}, \frac{\partial g_1}{\partial c^-_{N-1}} \\ \frac{\partial g_2}{\partial c^+_1}, \cdots, \frac{\partial g_2}{\partial c^+_{N-1}}, \frac{\partial g_2}{\partial c^-_1}, \cdots, \frac{\partial g_2}{\partial c^-_{N-1}} \end{bmatrix}$$



$$\frac{\partial g_2}{\partial c^{\pm}_n} = V_0 \left| \text{ume of } \bigcirc_{\text{at } n} \right. \times \underbrace{\Delta C_n}_{(=1)} \quad \forall n = \{3 : N_r - 3\}$$

$[m_0(\ ]$ $\quad [M^3]$ $\quad [m_0/M^3]$

$$= \frac{4}{3}\pi \left\{ \left(R^{\pm} \frac{n+1/2}{N_r} \right)^3 - \left(R^{\pm} \frac{n-1/2}{N_r} \right)^3 \right\}$$

$$= \frac{4(R^{\pm})^3 \pi}{3 N_r^3} \left\{ (n+\tfrac{1}{2})(n^2+n+\tfrac{1}{4}) - (n-\tfrac{1}{2})(n^2-n+\tfrac{1}{4}) \right\}$$

$$= \frac{4 R^3 \pi}{3 N_r^3} \left\{ \cancel{n^3} + \frac{1}{2}n^2 + n^2 + \cancel{\frac{1}{2}n} + \cancel{\frac{1}{4}n} + \frac{1}{8} - \left(\cancel{n^3} - \frac{1}{2}n^2 - \cancel{n^2} + \cancel{\frac{1}{2}n} + \cancel{\frac{1}{4}n} - \frac{1}{8} \right) \right\}$$

$$= \frac{4R^3\pi}{3N_r^3}\left(n^2+2n^2+\frac{1}{4}\right)$$

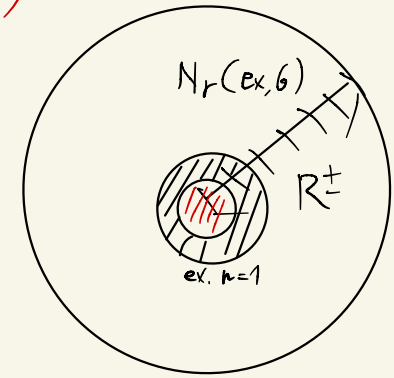
$$= \frac{4R^3\pi}{3N_r^3}\left(3n^2+\frac{1}{4}\right) \quad \forall n = \{3 : N_r - 3\}$$

For $n=1$:

$$\frac{\partial Q_2}{\partial C_1} = \frac{4\pi}{3}\left(\frac{R}{N_r}\right)^3\left(3(1)^2+\frac{1}{4}\right) + \frac{4}{3}\cdot\frac{4\pi}{3}\cdot\left(\frac{R}{N_r}\cdot\frac{1}{2}\right)^3$$

$$= \frac{4\pi}{3}\left(\frac{R}{N_r}\right)^3\left(\frac{13}{4}\right) + \frac{16\pi}{9}\left(\frac{R}{N_r}\right)^3\cdot\frac{1}{8}$$

$$= \frac{(39+2)}{9}\left(\frac{R}{N_r}\right)^3\pi = \frac{41}{9}\left(\frac{R}{N_r}\right)^3\pi$$



$$\frac{\partial Q_2}{\partial C_2} = \frac{4\pi}{3}\left(\frac{R}{N_c}\right)^3\left(3(2)^2+\frac{1}{4}\right) - \frac{1}{3}\cdot\frac{4\pi}{3}\left(\frac{R}{N_r}\cdot\frac{1}{2}\right)^3$$

$$= \frac{4\pi}{3}\left(\frac{R}{N_r}\right)^3\left(\frac{37}{4}\right) - \frac{14\pi}{9\cdot 2}\left(\frac{R}{N_r}\right)^3$$

$$= \pi\left(\frac{R}{N_r}\right)^3 \frac{37\times 3 - 1/2}{9} = \frac{221}{18}\pi\left(\frac{R}{N_r}\right)^3$$

$$\frac{\partial g_2}{\partial c_{N-2}} = \frac{4R^3\pi}{3Nr^3}\left(3(N-2)^2+\frac{1}{4}\right) - \frac{1}{3}\cdot\frac{4R^3\pi}{3Nr^3}\left(3N^2+\frac{1}{4}\right)$$

$$= \frac{4R^3\pi}{3Nr^3}\left(3(N^2-4N+4) - N^2 - \frac{1}{12}\right)$$

$$= \frac{4R^3\pi}{3Nr^3}\left(2N^2-12N+12-\frac{1}{12}\right)$$

$$\frac{\partial g_2}{\partial C_{N-1}} = \frac{4R^3\pi}{3Nr^3}\left(3(N+1)^2+\frac{1}{4}\right)+\frac{4}{3}\cdot\frac{4R^3\pi}{3Nr^3}\left(3N^2+\frac{1}{4}\right)$$

Kalman-Bucy filter

system

$$\dot{x} = Ax + Bu + L\omega$$

$(n\times 1)$ $(n\times n)\cdot(n\times 1)$ $(n\times m)\cdot(m\times 1)$ $(n\times s)\cdot(s\times 1)$

$$y = Cx + n$$

$(r\times 1)$ $(r\times n)\cdot(n\times 1)$ $(r\times 1)$

noise (zero-mean, gaussian)

$$E[\omega] = 0 \in \mathbb{R}^s$$

$$E[\omega(t)\omega(\tau)^T] = Q_c\,\delta(t-\tau)$$

Q_c: spectral density matrix

$$E[n] = 0$$

$$E[n(t)n(\tau)^T] = R_c\,\delta(t-\tau)$$

covariance

$$\Sigma_0 = E\{(x(0)-\hat{x}_0)(x(0)-\hat{x}_0)^T\} \in \mathbb{R}^{n\times n}$$

where $\hat{x}_0 = E[x(0)]$ is initial states

Covariance estimate

$$\dot{\Sigma} = A\Sigma + \Sigma A^T + \underbrace{L Q_c L^T}_{W \text{ (in Dong's note)}} - \Sigma C^T \overbrace{R_c^{-1}}^{N^{-1} \text{ (in Dong's note)}} C\Sigma$$

$(n\times n)\cdot(n\times n)$ $(n\times n)\cdot(n\times n)$ $(n\times s)(s\times s)(s\times n)$ $(n\times n)(n\times r)(r\times r)(r\times n)(n\times n)$

Filter gain

$$K_c = \Sigma C^T R_c^{-1}$$

State estimate

$$\dot{\hat{x}} = A\hat{x} + Bu + K_c(y - \hat{y})$$

Separation of electrode

$$x = \begin{bmatrix} c_p \\ \\ c_n \end{bmatrix} = \begin{bmatrix} c_{p,1} \\ \vdots \\ c_{p,N_r-1} \\ c_{n,1} \\ \vdots \\ c_{n,N_r-1} \end{bmatrix}$$

(plant)

$$\dot{x} = Ax + Bu$$

$$\Leftrightarrow \begin{bmatrix} \dot{\xi}_p \\ \dot{\xi}_n \end{bmatrix} = \begin{bmatrix} A_p & 0 \\ 0 & A_n \end{bmatrix} \begin{bmatrix} \xi_p \\ \xi_n \end{bmatrix} + \begin{bmatrix} B_p \\ B_n \end{bmatrix} u$$

$$\Leftrightarrow \quad \dot{\xi}_p = A_p \xi_p + B_p u$$

$$\dot{\xi}_n = A_n \xi_n + B_n u$$

(state estimator)

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - \hat{y})$$

$$\Leftrightarrow \begin{bmatrix} \dot{\hat{\xi}}_p \\ \dot{\hat{\xi}}_n \end{bmatrix} = \begin{bmatrix} A_p & 0 \\ 0 & A_n \end{bmatrix} \begin{bmatrix} \hat{\xi}_p \\ \hat{\xi}_n \end{bmatrix} + \begin{bmatrix} B_p \\ B_n \end{bmatrix} u + \begin{bmatrix} L_p \\ L_n \end{bmatrix} (y - \hat{y})$$

$$\Leftrightarrow \quad \dot{\hat{\xi}}_p = A_p \hat{\xi}_p + B_p u + L_p (y - \hat{y})$$

$$\dot{\hat{\xi}}_n = A_n \hat{\xi}_n + B_n u + L_n (y - \hat{y})$$

(covariance estimator) → cannot separate two electrodes

$$\dot{\Sigma} = \Sigma \cdot A^T + A\Sigma + W - \Sigma \cdot H^T \cdot N^{-1} \cdot H \cdot \Sigma$$

$$\Leftrightarrow \begin{bmatrix} \dot{\Sigma}_p & \dot{\Sigma}_{pn} \\ \dot{\Sigma}_{np} & \dot{\Sigma}_n \end{bmatrix} = \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix} \begin{bmatrix} A_p & 0 \\ 0 & A_n \end{bmatrix} + \begin{bmatrix} A_p & 0 \\ 0 & A_n \end{bmatrix} \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix}$$

$$+ \begin{bmatrix} W_p & W_{pn} \\ W_{np} & W_n \end{bmatrix} - \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix} \begin{bmatrix} H_p^T \\ H_n^T \end{bmatrix} N^{-1} \begin{bmatrix} H_p & H_n \end{bmatrix} \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix}$$

$$\Leftarrow \begin{bmatrix} \dot{\Sigma}_p & \dot{\Sigma}_{pn} \\ \dot{\Sigma}_{np} & \dot{\Sigma}_n \end{bmatrix} = \begin{bmatrix} \Sigma_p A_p & \Sigma_{pn} A_n \\ \Sigma_{np} A_p & \Sigma_n A_n \end{bmatrix} + \begin{bmatrix} A_p \Sigma_p & A_p \Sigma_{pn} \\ A_n \Sigma_{np} & A_n \Sigma_n \end{bmatrix} + \begin{bmatrix} W_p & W_{pn} \\ W_{np} & W_n \end{bmatrix}$$

$$- \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix} \begin{bmatrix} H_p^T N^{-1} \\ H_n^T N^{-1} \end{bmatrix} \begin{bmatrix} H_p & H_n \end{bmatrix} \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix} \begin{bmatrix} H_p^T N^{-1} H_p & H_p^T N^{-1} H_n \\ H_n^T N^{-1} H_p & H_n^T N^{-1} H_n \end{bmatrix} \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix}$$

$$= \begin{bmatrix} \Sigma_p H_p^T N^{-1} H_p + \Sigma_{pn} H_n^T N^{-1} H_p & \Sigma_p H_p^T N^{-1} H_n + \Sigma_{pn} H_n^T N^{-1} H_n \\ \Sigma_{np} H_p^T N^{-1} H_p + \Sigma_n H_n^T N^{-1} H_p & \Sigma_{np} H_p^T N^{-1} H_n + \Sigma_n H_n^T N^{-1} H_n \end{bmatrix} \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix}$$

(gain) → cannot separate (calculation)

$$L = \Sigma H^T N^{-1}$$

$$\Leftrightarrow \begin{bmatrix} L_p \\ L_n \end{bmatrix} = \begin{bmatrix} \Sigma_p & \Sigma_{pn} \\ \Sigma_{np} & \Sigma_n \end{bmatrix} \begin{bmatrix} H_p^T \\ H_n^T \end{bmatrix} N^{-1}$$

$$= \begin{bmatrix} \Sigma_p H_p^T + \Sigma_{pn} H_n^T \\ \Sigma_{np} H_p^T + \Sigma_n H_n^T \end{bmatrix} N^{-1}$$

$$\dot{C}^{\pm}_{s,N-1} = \frac{D^{\pm}_s}{(R^{\pm}_s)^2(\Delta\bar{r})^2}\left(\frac{N-2}{N-1}C^{\pm}_{s,N-2} - 2C^{\pm}_{s,N-1} + \frac{N}{N-1}C^{\pm}_{s,N}\right)$$

$$\frac{\partial C^{\pm}_s}{\partial \bar{r}}(1,t) = \frac{3C^{\pm}_{s,N} - 4C^{\pm}_{s,N-1} + C^{\pm}_{s,N-2}}{2\Delta\bar{r}\cdot R^{\pm}_s} = \frac{\pm I(t)}{D^{\pm}_s F\alpha^{\pm} A^{\pm} L^{\pm}}$$

$$\frac{\partial(\dot{C}^{\pm}_{s\,N-1})}{\partial I} = \frac{\partial\left(\frac{N-2}{N-1}C^{\pm}_{s,N-2} - 2C^{\pm}_{s\,N-1} + \frac{N}{N-1}C^{\pm}_{s,N}\right)}{\partial I}$$

$$= \frac{N-2}{N-1}\frac{\partial(C^{\pm}_{s,N-2})}{\partial I} - 2\frac{\partial(C^{\pm}_{s,N-1})}{\partial I} + \frac{N}{N-1}\frac{\partial(C^{\pm}_{s,N})}{\partial I}$$

$$= \pm\frac{2\Delta\bar{r}\,R^{\pm}_s}{D^{\pm}_s F\alpha^{\pm} A^{\pm} L^{\pm}}\left(\frac{N-2}{N-1}1 - 2\cdot\frac{1}{4} + \frac{N}{N-1}\cdot\frac{1}{3}\right)$$

$$\frac{\partial(\dot{C}^{\pm}_{s,N-2})}{\partial I} = \frac{\partial\left(\frac{N-3}{N-2}C^{\pm}_{s,N-3} - 2C^{\pm}_{s,N-2} + \frac{N-1}{N-2}C^{\pm}_{s,N-1}\right)}{\partial I}$$

$$= \pm \frac{2\Delta\bar{r}\, R_s^{\pm}}{P_s^{\pm} \mp a^{\pm} A^{\pm} L^{\pm}} \left(-2 + \frac{N-1}{N-2}\left(-\frac{1}{4}\right) \right)$$

$$\frac{\partial \left(\dot{C}_{s,N-3}^{\pm} \right)}{\partial I} = \frac{\partial \left(\frac{N-4}{N-3} C_{s,N-4}^{\pm} - 2 C_{s,N-3}^{\pm} + \frac{N-2}{N-3} C_{s,N-2}^{\pm} \right)}{\partial I}$$

$$= \pm \frac{2\Delta\bar{r}\, R_s^{\pm}}{P_s^{\pm} \mp a^{\pm} A^{\pm} L^{\pm}} \left(\frac{N-2}{N-3} \right).$$

KF with constraint

$$x_k = F_{k,k-1} x_{k-1} + u_{k,k-1} \qquad u_{k,k-1} \sim N(0, Q_{k,k-1})$$

$$z_k = H_k x_k + v_k \qquad v_k \sim N(0, R_k)$$

state prediction error : $\tilde{x}_{k|k-1} = x_k - \hat{x}_{k|k-1}$